Ex.no:3 Packet tracer tool Installation & User Interface Overview

Aim:- To study the packet tracer tool installation & user interface overview.

To understand environment of CISCO PACKET TRACER to design simple network.

Introduction:-

A simulator as the name suggests, simulates network devices and its environment. Packet Tracer is an existing network design, simulation & modelling tool.

1. It allows you to model complex systems without the need for dedicated equipment.
2. It helps you to practice your network configuration and troubleshooting skills via computer or an android or ios based mobile device.
3. It is available for both the Linux & Windows desktop environment
4. Protocols in Packet Tracer are coded to work & behave in the same way as they would on real hardware.

Installing Packet Tracer:-

Installation in Windows is pretty simple & straightforward. the setup comes in a named PacketTracer6_0_1_setup. Open this file to begin the setup wizard, accept agreement, choose a location & start the installation.

User Interface Overview:-

The layout of Packet Tracer is divided into several components, The components of Packet Tracer interface are as follows: match the numbering with explanations.

1. Menu bar
2. Main Toolbar
3. Logical / Physical workspace tabs
4. Workspace
5. Common tools bar
6. Real-time / Simulation tabs
7. Network component box
8. User-created packet box.

d). Analyse the behaviour of network devices using CISCO PACKET TRACER simulator:-

1. From the network component box, click & drag & drop the below component a). 4 Generic PCs & One HUB
   b). 4 Generic PCs & One Switch
2. Click on Connections:
   a). Click on copper Straight-Through cable.
   b). Select one PC & connect it to HUB.
   c). Similarly connect 4 PCs to the switch
3. Click on the PCs connected to hub, go to desktop tab, click on IP configuration & enter an IP address & subnet mask.
4. Observe the flow of PDU from source PC to destination PC by selecting the Realtime mode of simulation.
5. Repeat step #3 to step #5 for the PCs connected to the switch.
6. Observe how HUB & switch are forwarding the PDU & write your observation & conclusion about the behaviour of switch & HUB.

Student Observation:-

a). From your observation write down the behaviour of switch & HUB in terms of forwarding the packets received & by them.

Ans. HUB (Broadcasting)
- Operates at Layer 1 (Physical)
- Cannot read MAC address
- Lacks privacy & efficiency

SWITCH (Unicasting):-
- Operates at Layer 2 (Data link)
- Reads MAC address
- Improves network security & eliminates data collision.

b). Find out the network topology implemented in your college & draw and label that topology in your observation book. End devices: Computers laptops & printers

Ans Internet Cloud: Main external source
Core Routes: The central campus gateway
Distribution switches: Main building hubs
Access switches: Individual classroom connection







